

Project Vitriol – NEA Evaluation

The final implemented system successfully meets the functional core objectives defined during the Analysis Section. Core features including map creation with defined dimensions, layered tile editing, collision handling, event placement, JSON persistence, and runtime playback were implemented and validated through structured subsystem and regression testing. The separation between editor logic, runtime behaviour, and shared data models proved effective in maintaining structural ability as new features continued to be introduced. The use of passive DTOs and a single serialisation pipeline ensured that map files remained consistent and portable through development, reducing the risk of format drift or data corruption.

From a performance perspective, the viewport-only rendering model achieved the intended purpose: large maps remained usable without excessive UI lag, and repeated rendering operations did not introduce visible degradation. The command-based undo/redo system behaved consistently across tile, collision, and event edits, confirming that the architectural decisions made in the Design Section translated effectively into implementation. User testing further indicated that the primary workflows: map creation, tile painting, and runtime interaction, were intuitive, with only minor discoverability issues in the event editing interface.

However, there are still several limitations to this project. The event command system, while flexible, still requires careful manual configuration and lacks higher-level tooling such as validation previews or visual scripting aids. Additionally, although performance remained stable during observational testing, more formal profiling tools could provide deeper insight into memory usage and rendering cost under extreme conditions. Usability could also be improved by refining tool discoverability and clarifying autosave behaviour to reduce potential user confusion.

Overall, the system satisfies the stated requirements, and demonstrates functional completeness, structural stability, and practical usability. While there is scope for refinement and expansion, particularly in terms of usability and tooling depth, the project achieves its intended goals within the defined scope and technical constraints.